Yuchen Hou

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Summary

Master student at the National University of Singapore, focusing on Embodied AI and autonomous robotic decision-making.

Education

M.Sc. **National University of Singapore**, Electrical Engineering — Singapore, Aug 2025 – present

- Core member of the NUS Advanced Control Lab, focusing on Embodied AI and autonomous robotic decision-making.
- Research on perception and planning for aerial robots based on Vision-Language Navigation (VLN), and end-to-end Reinforcement Learning in complex 3D environments.

B.Eng. **Nanjing University of Aeronautics and Astronautics**, Electronic and Communication Engineering — Nanjing, China, Sept 2021 – July 2025

- GPA: 88/100
- Key Courses: Advanced Mathematics (96), Probability Theory (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99).

Experience

Honor Device Co., Ltd., VLA Algorithm Engineer (Intern) — Shanghai, China, June 2026 – present, 3 months

- Working on Honor's self-developed Omega humanoid robot: teleoperation data collection, replay, training, and on-robot deployment.
- Conducting online RL algorithm research for Vision-Language-Action (VLA) models on custom manipulation tasks.
- Implemented smooth human-teleoperator takeover during autonomous rollouts, including the autonomy-teleoperation transition state machine.
- Built the intervention-driven online-learning data pipeline: on-robot data recording, full local disk persistence, and automatic upload to the development server.
- Closed the loop over a distributed learning architecture: onboard AGX/NX boards (topic publishing, command execution, data collection), an x86 host as the Actor (model inference, streaming actions to the robot), and a development server as the Learner (data aggregation, training, checkpoint dispatch)
- a complete online SFT / online RL pipeline.
- Researching RECAP-style (pi-0.6) offline RL combined with noise-space fast adaptation (DSRL-like): learning corrective noise from human-intervention data to rapidly imitate takeover behaviors before a new checkpoint finishes training, substantially raising success rates on tasks uncovered by offline training, and further adapting the algorithm to loco-manipulation tasks.

Huawei Norbert Wiener Research Center, Research Intern — Singapore, Mar 2026 – May 2026, 3 months

- Developed backend deep learning models for short-range radar signal processing.
- Conducted research on downstream tasks including human pose estimation and indoor mapping/SLAM from radar data.

Movel.AI, Robotics Software Intern — Singapore — Dec 2025 – Jan 2026 — 2 months

- Validated autonomous navigation and control systems for forklift robots in industrial environments.
- Conducted on-site factory deployment and testing in real warehouse settings.
- Integrated robotics software with fleet management systems.
- Tech Stack: ROS2, Navigation Stack, Sensor Integration.

Awards

Second-class Scholarship: Nanjing University of Aeronautics and Astronautics (2024)

Third-class Scholarship: Nanjing University of Aeronautics and Astronautics (2021 – 2023)

Skills

Programming & Systems: Python (ROS 2, PyTorch, OpenCV), Linux (Ubuntu), Shell scripting

AI Frameworks & Simulation: PyTorch (DRL policies), Habitat, AI2-THOR (Complex 3D environments)

Robotics Development: ROS2 (rclpy, Nav2), Sensor Fusion, Lidar/RGB-D Integration, SLAM (Cartographer)

Research Capabilities: Sim-to-Real Transfer, Embodied AI Architectures, RL Algorithms, Academic Paper Writing, Tracking CVPR/IROS/ICRA

Languages

English: Proficient — IELTS 7.0 (R 8.5, W 7.5, L 6.5, S 6.0)

Mandarin: Native speaker